THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Half of the supplement of 60° is: 1

(a) 60° **(b)** 50° **(c)** 70° **(d)** 80°

2. The angle bisector of a 110° angle makes each part: 1

(a) 65° **(b)** 75° **(c)** 55° **(d)** 45°

3. A transversal makes corresponding angles of 70° and 75°. The two lines are: 1

(a) perpendicular **(b)** equal **(c)** not parallel **(d)** parallel

4. A transversal makes co-interior angles in the ratio $2:3$ on parallel lines. The smaller angle is: 1

(a) 62° **(b)** 72° **(c)** 82° **(d)** 92°

5. If $(x+5)^\circ$ and $(2x-14)^\circ$ are supplementary, x equals: 1

(a) 73 **(b)** 58 **(c)** 68 **(d)** 63

6. If $l \parallel m$ and two co-interior angles are 99° and $(x)^\circ$, then $x =$ 1

(a) 101° **(b)** 91° **(c)** 81° **(d)** 71°

7. The angle which is 8 times its complement is: 1

(a) 70° **(b)** 100° **(c)** 80° **(d)** 90°

8. Half of the supplement of 100° is: 1

(a) 60° **(b)** 50° **(c)** 30° **(d)** 40°

9. The angle bisector of a 130° angle makes each part: 1

(a) 55° **(b)** 75° **(c)** 85° **(d)** 65°

10. A transversal makes co-interior angles of 110° and 70°. The two lines are: 1

(a) parallel **(b)** perpendicular **(c)** cannot say **(d)** not parallel

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find x if $(3x-6)^\circ$ and $(x+14)^\circ$ are vertically opposite angles. 2

12. Is $l \parallel m$ if a transversal gives corresponding angles 110° and 105°? Give reason. 2

13. Find the angle that is 60° less than its supplement. 2

14. Find x if $(3x-12)^\circ$ and $(x+40)^\circ$ are vertically opposite angles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A transversal cuts parallel lines. Interior angles on the same side are $(2x + 15)^\circ$ and $(3x + 10)^\circ$. Find x and the two angles. **3**

16. In a pair of intersecting lines, the angles are $(x + 12)^\circ$, $(2x - 3)^\circ$ forming a linear pair. Find x and the vertically opposite angles. **3**

17. An angle is 30° more than twice its complement. Find the angle and its complement. **3**

OR

A transversal cuts parallel lines. Interior angles on the same side are $(2x + 20)^\circ$ and $(3x + 25)^\circ$. Find x and the two angles. **3**

18. In a pair of intersecting lines, the angles are $(x + 24)^\circ$, $(2x - 12)^\circ$ forming a linear pair. Find x and the vertically opposite angles. **3**

OR

An angle is 45° more than twice its complement. Find the angle and its complement. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Lines $l \parallel m$, transversal t. $\angle 1 = (2x + 5)^\circ$, $\angle 2 = (3x - 25)^\circ$ are alternate interior angles; $\angle 3$ is the co-interior angle of $\angle 1$. Find x, $\angle 1$, $\angle 2$ and $\angle 3$. **5**

OR

Around a point, angles are x°, $2x^\circ$, $3x^\circ$ and 90°. Find x and each angle, and say which are acute, right or obtuse. **5**

20. Lines $l \parallel m$, transversal t. $\angle 1 = (2x + 20)^\circ$, $\angle 2 = (3x - 15)^\circ$ are alternate interior angles; $\angle 3$ is the co-interior angle of $\angle 1$. Find x, $\angle 1$, $\angle 2$ and $\angle 3$. **5**

OR

Around a point, angles are x°, $2x^\circ$, $3x^\circ$ and 120°. Find x and each angle, and say which are acute, right or obtuse. **5**

END OF QUESTION PAPER — Total: 40 Marks

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